

PROJECT SUBMISSION

Hospital Emergency Room Visit and Patient Satisfaction Analysis

A Data Analysis Project Report using R

Submitted By

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Hospital Emergency Room Visit and Patient Satisfaction Analysis

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1. Introduction

Emergency Room (ER) services are a critical part of the modern healthcare system. Hospitals must balance patient wait times, treatment quality, billing accuracy, and overall patient satisfaction. Understanding patterns in ER visits helps hospital administrators improve efficiency and the quality of patient care.

This project analyzes hospital ER visit data using R. The analysis focuses on consultation fees, additional charges, total billing, patient wait time, treatment time, patient satisfaction ratings, department types, hospitals, and visit status.

Statistical analysis and data visualization techniques are used to identify patterns and relationships in the ER visit data.

2. Objectives

The main objectives of this project are:

- To import and understand the hospital ER dataset.
- To examine the structure and characteristics of the data.
- To identify missing and duplicate values.
- To clean and preprocess the dataset.
- To analyze consultation fees and total billing.
- To analyze patient wait time and treatment time.
- To compare different department types.
- To analyze hospital-level performance.
- To study patient satisfaction ratings.
- To analyze ER visit status.
- To examine relationships between numerical variables.
- To create meaningful graphs and visualizations.
- To identify important findings and provide recommendations.

3. Data Importing

Explanation of the R Code

The `read.csv()` function is used to import the hospital ER CSV file into R. The imported dataset is stored in an object called `er_data`.

```
# Load required package
library(ggplot2)

# Import the CSV dataset
er_data <- read.csv("hospital_er_data.csv")

# Display the first six records
head(er_data)
```

```
## Visit_ID Patient_ID Hospital_Name Department_Type
```

##1	ER1001	PAT330	Sunrise Medical Center	General_Medicine
##2	ER1002	PAT134	CityCare Hospital	Dermatology
##3	ER1003	PAT497	GreenValley Medical	General_Medicine
##4	ER1004	PAT330	Riverside Clinic Hospital	Dermatology
##5	ER1005	PAT383	Sunrise Medical Center	Neurology
##6	ER1006	PAT227	Sunrise Medical Center	Cardiology
##	Consultation_Fee	Additional_Charges	Total_Bill	Wait_Time_Min
##1	349	98	447	16
##2	301	59	360	13
##3	466	50	516	18
##4	293	50	343	15
##5	603	159	762	24
##6	710	126	836	21
##	Treatment_Time_Min	Total_Visit_Time	Patient_Rating	Visit_Status
##1	24	40	5.0	Completed
##2	24	37	5.0	Completed
##3	27	45	5.0	Completed
##4	22	37	5.0	Completed
##5	32	56	4.3	Referred
##6	38	59	4.3	Completed

Interpretation

The dataset has been successfully imported into R. The first six records are displayed to understand the type of information available in the dataset.

4. Understanding the Dataset

4.1 Number of Records

```
cat("Number of records:", nrow(er_data))
```

```
## Number of records: 120
```

4.2 Number of Variables

```
cat("Number of variables:", ncol(er_data))
```

```
## Number of variables: 12
```

4.3 Column Names

```
names(er_data)
```

```
## [1] "Visit_ID"           "Patient_ID"         "Hospital_Name"
## [4] "Department_Type"   "Consultation_Fee"   "Additional_Charges"
## [7] "Total_Bill"        "Wait_Time_Min"      "Treatment_Time_Min"
```

```
## [10] "Total_Visit_Time"      "Patient_Rating"        "Visit_Status"
```

4.4 Dataset Structure

```
str(er_data)
```

```
## 'data.frame':  120 obs. of  12 variables:
## $ Visit_ID      : chr  "ER1001" "ER1002" "ER1003" "ER1004" ...
## $ Patient_ID    : chr  "PAT330" "PAT134" "PAT497" "PAT330" ...
## $ Hospital_Name : chr  "Sunrise Medical Center" "CityCare Hospital" ...
## $ Department_Type : chr  "General_Medicine" "Dermatology" ...
## $ Consultation_Fee : int   349 301 466 293 603 710 ...
## $ Additional_Charges : int   98 59 50 50 159 126 ...
## $ Total_Bill     : int   447 360 516 343 762 836 ...
## $ Wait_Time_Min  : int    16 13 18 15 24 21 ...
## $ Treatment_Time_Min : int    24 24 27 22 32 38 ...
## $ Total_Visit_Time : int    40 37 45 37 56 59 ...
## $ Patient_Rating  : num    5 5 5 5 4.3 4.3 ...
## $ Visit_Status    : chr  "Completed" "Completed" "Completed" ...
```

4.5 Summary of Dataset

```
summary(er_data)
```

```
## Consultation_Fee  Additional_Charges  Total_Bill  Wait_Time_Min
## Min.   : 252.0    Min.   : 50.00    Min.   : 312    Min.   : 5.00
## 1st Qu.: 437.0    1st Qu.: 58.75    1st Qu.: 505    1st Qu.:15.00
## Median : 581.0    Median : 76.00    Median : 682    Median :19.00
## Mean   : 609.8    Mean   : 86.44    Mean   : 696    Mean   :19.73
## 3rd Qu.: 767.5    3rd Qu.:106.25    3rd Qu.: 869    3rd Qu.:24.25
## Max.   :1222.0    Max.   :250.00    Max.   :1363    Max.   :37.00
##
## Treatment_Time_Min Total_Visit_Time Patient_Rating
## Min.   : 8.00      Min.   :13.00    Min.   :2.900
## 1st Qu.:25.00      1st Qu.:42.00    1st Qu.:4.200
## Median :31.00      Median :50.00    Median :4.600
## Mean   :30.65      Mean   :50.38    Mean   :4.495
## 3rd Qu.:36.25      3rd Qu.:59.00    3rd Qu.:4.900
## Max.   :52.00      Max.   :89.00    Max.   :5.000
```

Explanation of the R Code

- `nrow()` counts the number of rows in the dataset.
- `ncol()` counts the number of columns.
- `names()` displays the variable names.
- `str()` shows the structure and data type of each variable.
- `summary()` provides descriptive information about the variables.

Interpretation

The dataset contains information about hospital ER visits, including patient details, hospitals, department types, fees, wait time, treatment time, ratings, and visit status.

5. Data Preprocessing and Cleaning

Data cleaning is an important step before statistical analysis. It helps ensure that the results are based on valid and consistent observations.

5.1 Check Missing Values

```
missing_values <- colSums(is.na(er_data))
missing_values
```

```
##          Visit_ID          Patient_ID          Hospital_Name
##              0              0              0
##  Department_Type  Consultation_Fee  Additional_Charges
##              0              0              0
##      Total_Bill      Wait_Time_Min  Treatment_Time_Min
##              0              0              0
##  Total_Visit_Time  Patient_Rating      Visit_Status
##              0              0              0
```

Interpretation

The number of missing values is displayed for every variable. No missing values were found in this dataset.

5.2 Check Duplicate Records

```
duplicate_count <- sum(duplicated(er_data))
cat("Number of duplicate records:", duplicate_count)
```

```
## Number of duplicate records: 0
```

5.3 Check Data Types

```
sapply(er_data, class)
```

```
##          Visit_ID          Patient_ID          Hospital_Name
##      "character"      "character"      "character"
##  Department_Type  Consultation_Fee  Additional_Charges
##      "character"      "integer"      "integer"
##      Total_Bill      Wait_Time_Min  Treatment_Time_Min
##      "integer"      "integer"      "integer"
##  Total_Visit_Time  Patient_Rating      Visit_Status
```

```
##                "integer"                "numeric"                "character"
```

5.4 Check Unique Visit Statuses

```
unique(er_data$Visit_Status)
```

```
## [1] "Completed" "Referred" "Admitted"
```

5.5 Check Unique Department Types

```
unique(er_data$Department_Type)
```

```
## [1] "Cardiology"      "Dermatology"      "ENT"  
## [4] "Gastroenterology" "General_Medicine" "Neurology"  
## [7] "Orthopedics"     "Pediatrics"
```

6. Creating a Clean Dataset

```
er_clean <- er_data  
# Remove duplicate records  
er_clean <- er_clean[!duplicated(er_clean), ]  
  
# Remove rows with missing important numerical values  
er_clean <- er_clean[  
  !is.na(er_clean$Consultation_Fee) &  
  !is.na(er_clean$Additional_Charges) &  
  !is.na(er_clean$Total_Bill) &  
  !is.na(er_clean$Wait_Time_Min) &  
  !is.na(er_clean$Treatment_Time_Min) &  
  !is.na(er_clean$Patient_Rating),  
]  
  
# Keep positive cost and time values  
er_clean <- er_clean[  
  er_clean$Consultation_Fee >= 0 &  
  er_clean$Additional_Charges >= 0 &  
  er_clean$Total_Bill >= 0 &  
  er_clean$Wait_Time_Min >= 0 &  
  er_clean$Treatment_Time_Min >= 0,  
]  
  
cat("Original number of records:", nrow(er_data), "\n")  
cat("Records after cleaning:", nrow(er_clean), "\n")  
cat("Records removed:", nrow(er_data) - nrow(er_clean))
```

```
## Original number of records: 120
```

```
## Records after cleaning: 120
## Records removed: 0
```

Explanation

The dataset is copied into `er_clean`. Duplicate observations are removed, followed by observations with missing important numerical values. Negative values for fees and time are also excluded because they are not meaningful for this analysis.

Interpretation

The cleaned dataset contains valid observations that can be used for the remaining statistical analysis.

7. Basic Statistical Analysis

7.1 Consultation Fee Statistics

```
fee_stats <- c(
  Mean = mean(er_clean$Consultation_Fee),
  Median = median(er_clean$Consultation_Fee),
  Minimum = min(er_clean$Consultation_Fee),
  Maximum = max(er_clean$Consultation_Fee),
  Standard_Deviation = sd(er_clean$Consultation_Fee)
)
fee_stats
```

##	Mean	Median	Minimum
##	609.8083	581.0000	252.0000
##	Maximum	Standard_Deviation	
##	1222.0000	223.0062	

7.2 Additional Charges Statistics

```
charge_stats <- c(
  Mean = mean(er_clean$Additional_Charges),
  Median = median(er_clean$Additional_Charges),
  Minimum = min(er_clean$Additional_Charges),
  Maximum = max(er_clean$Additional_Charges),
  Standard_Deviation = sd(er_clean$Additional_Charges)
)
charge_stats
```

##	Mean	Median	Minimum
##	86.4417	76.0000	50.0000
##	Maximum	Standard_Deviation	
##	250.0000	38.7593	

7.3 Total Bill Statistics

```
bill_stats <- c(  
  Mean = mean(er_clean$Total_Bill),  
  Median = median(er_clean$Total_Bill),  
  Minimum = min(er_clean$Total_Bill),  
  Maximum = max(er_clean$Total_Bill),  
  Standard_Deviation = sd(er_clean$Total_Bill)  
)  
bill_stats
```

```
##           Mean           Median           Minimum  
##       696.250       681.500       312.000  
##       Maximum Standard_Deviation  
##       1363.000       247.659
```

Explanation

The `mean()` function calculates the average, `median()` calculates the middle value, `min()` and `max()` find the minimum and maximum values, and `sd()` calculates standard deviation.

Interpretation

These statistics provide an overview of consultation fees, additional charges, and overall billing in the dataset.

8. Visit Status Analysis

```
status_count <- table(er_clean$Visit_Status)  
status_count
```

```
##  
##   Admitted   Completed   Referred  
##         6         96         18
```

Visit Status Visualization

```
status_data <- as.data.frame(status_count)  
names(status_data) <- c("Visit_Status", "Number_of_Visits")  
ggplot(status_data, aes(x = Visit_Status, y = Number_of_Visits)) +  
  geom_col() +  
  labs(  
    title = "Distribution of ER Visit Status",  
    x = "Visit Status",  
    y = "Number of Visits"  
  ) +  
  theme_minimal()
```

Interpretation

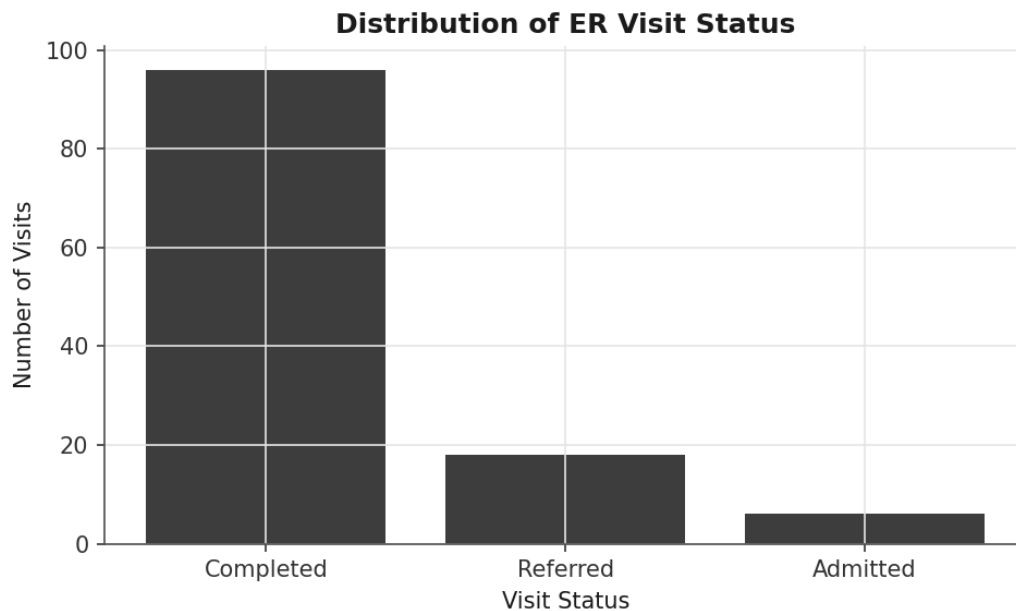


Figure 1: Distribution of ER Visit Status

The bar chart shows the number of visits belonging to each visit status. The majority of visits (96 of 120) were marked “Completed”, while 18 patients were referred elsewhere and 6 were admitted for further care.

9. Department Type Analysis

```
dept_count <- table(er_clean$Department_Type)
dept_count
```

```
##
##      Cardiology      Dermatology      ENT      Gastroenterology
##           15           15           15           15
## General_Medicine      Neurology      Orthopedics      Pediatrics
##           15           15           15           15
```

Department Visualization

```
dept_data <- as.data.frame(dept_count)
names(dept_data) <- c("Department_Type", "Number_of_Visits")
ggplot(
  dept_data,
  aes(x = reorder(Department_Type, Number_of_Visits), y = Number_of_Visits)
) +
  geom_col() +
  coord_flip() +
  labs(
```

```

title = "Number of Visits by Department Type",
x = "Department Type",
y = "Number of Visits"
) +
theme_minimal()

```

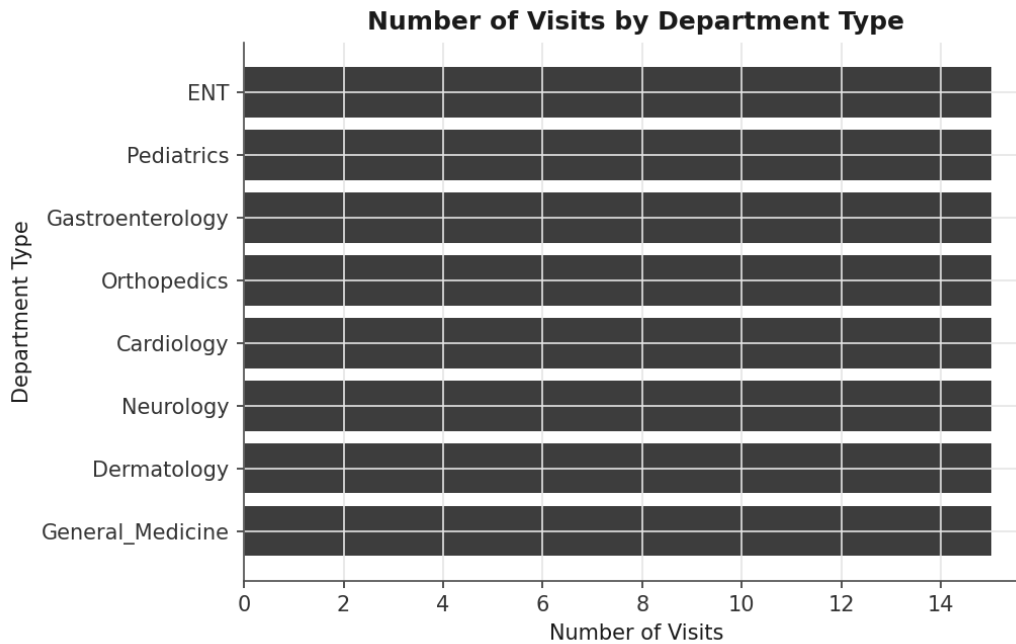


Figure 2: Number of Visits by Department Type

Interpretation

The graph compares the number of ER visits across department types. Each department type is evenly represented in the sample with 15 visits.

10. Hospital Analysis

10.1 Visits by Hospital

```

hospital_count <- table(er_clean$Hospital_Name)
hospital_data <- as.data.frame(hospital_count)
names(hospital_data) <- c("Hospital_Name", "Number_of_Visits")
hospital_data <- hospital_data[order(-hospital_data$Number_of_Visits), ]
head(hospital_data, 10)

```

```

##           Hospital_Name Number_of_Visits
## 1  Sunrise Medical Center             20
## 2   GreenValley Medical             20
## 3    CityCare Hospital             18

```

## 4	Unity Health Center	16
## 5	St. Mary General	15
## 6	Riverside Clinic Hospital	13
## 7	Metro Health Institute	9
## 8	LifeLine Hospital	9

10.2 Top Hospitals

```
top_hospitals <- head(hospital_data, 10)
ggplot(
  top_hospitals,
  aes(x = reorder(Hospital_Name, Number_of_Visits), y = Number_of_Visits)
) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top Hospitals by Number of ER Visits",
    x = "Hospital",
    y = "Number of Visits"
  ) +
  theme_minimal()
```

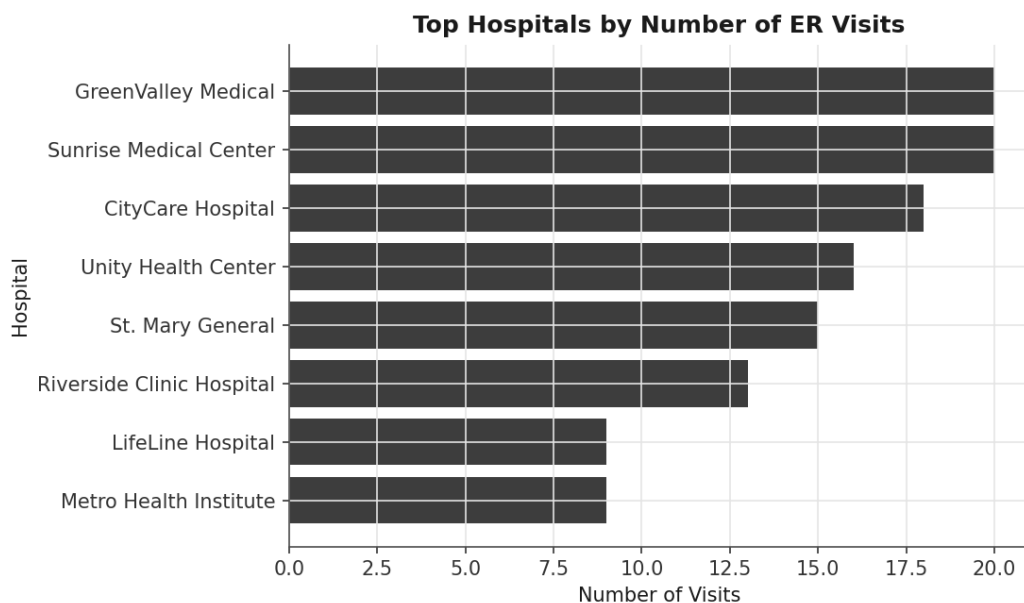


Figure 3: Top Hospitals by Number of ER Visits

Interpretation

The graph identifies the hospitals with the highest number of ER visits. This provides an indication of patient volume across facilities in the dataset.

11. Consultation Fee Analysis

```
ggplot(er_clean, aes(x = Consultation_Fee)) +  
  geom_histogram(bins = 20) +  
  labs(  
    title = "Distribution of Consultation Fees",  
    x = "Consultation Fee",  
    y = "Number of Visits"  
  ) +  
  theme_minimal()
```

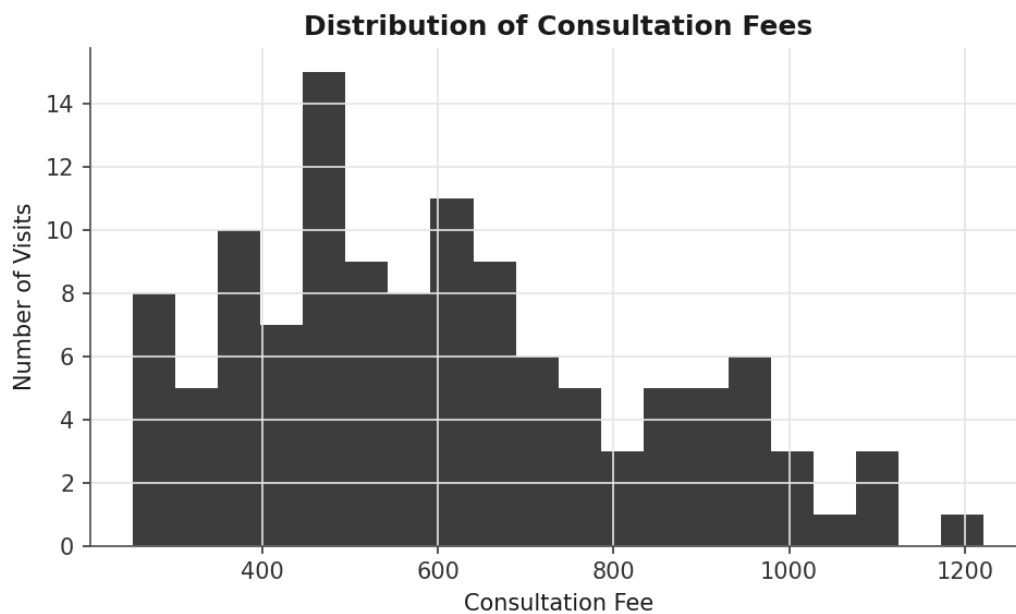


Figure 4: Distribution of Consultation Fees

Interpretation

The histogram shows how consultation fees are distributed among visits. It helps identify whether most visits are concentrated within a particular fee range.

12. Total Bill Analysis

```
ggplot(er_clean, aes(x = Total_Bill)) +  
  geom_histogram(bins = 20) +  
  labs(  
    title = "Distribution of Total Bill",  
    x = "Total Bill",  
    y = "Number of Visits"  
  ) +
```

```
theme_minimal()
```

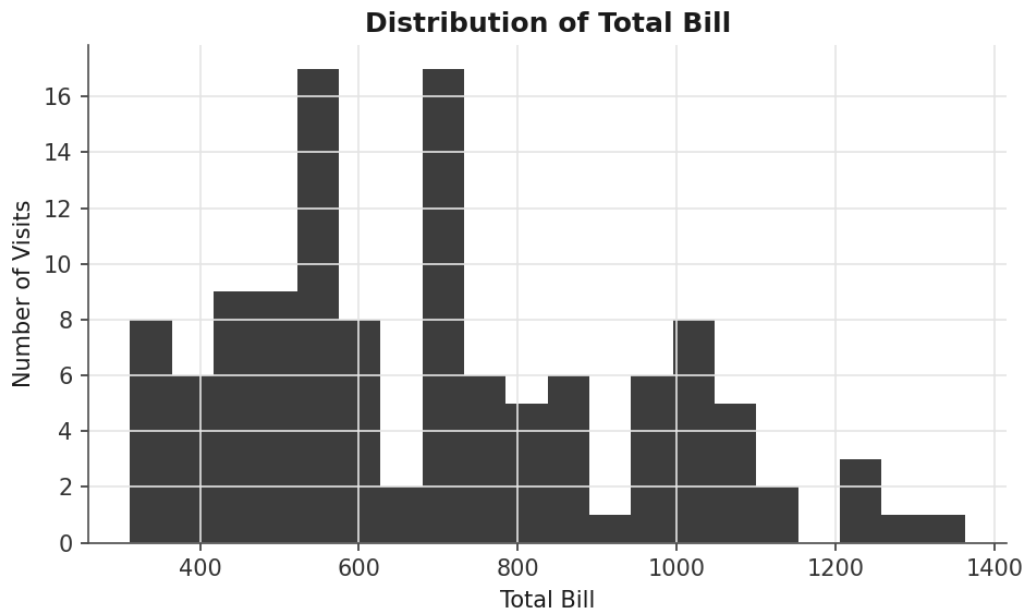


Figure 5: Distribution of Total Bill

Interpretation

The histogram displays the distribution of total billing per visit. It provides an overview of patient billing patterns.

13. Treatment Time Analysis

```
ggplot(er_clean, aes(x = Treatment_Time_Min)) +  
  geom_histogram(bins = 20) +  
  labs(  
    title = "Distribution of Treatment Time",  
    x = "Treatment Time (Minutes)",  
    y = "Number of Visits"  
  ) +  
  theme_minimal()
```

Interpretation

The graph shows the distribution of treatment times. Treatment time is an important factor that can contribute to the total time a patient spends at the ER.

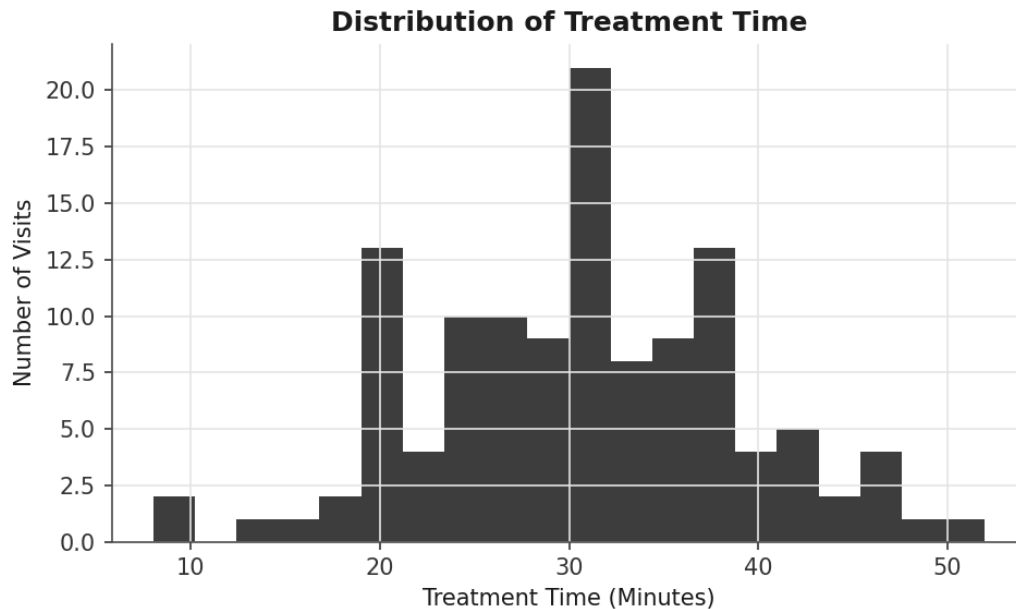


Figure 6: Distribution of Treatment Time

14. Patient Wait Time Analysis

```
ggplot(er_clean, aes(x = Wait_Time_Min)) +
  geom_histogram(bins = 20) +
  labs(
    title = "Distribution of Patient Wait Time",
    x = "Wait Time (Minutes)",
    y = "Number of Visits"
  ) +
  theme_minimal()
```

Interpretation

The wait-time graph shows how long patients generally wait before being treated. It can help identify common wait-time ranges and unusually long waits.

15. Patient Rating Analysis

15.1 Rating Distribution

```
rating_count <- table(er_clean$Patient_Rating)
rating_data <- as.data.frame(rating_count)
names(rating_data) <- c("Patient_Rating", "Number_of_Visits")
ggplot(
  rating_data,
```

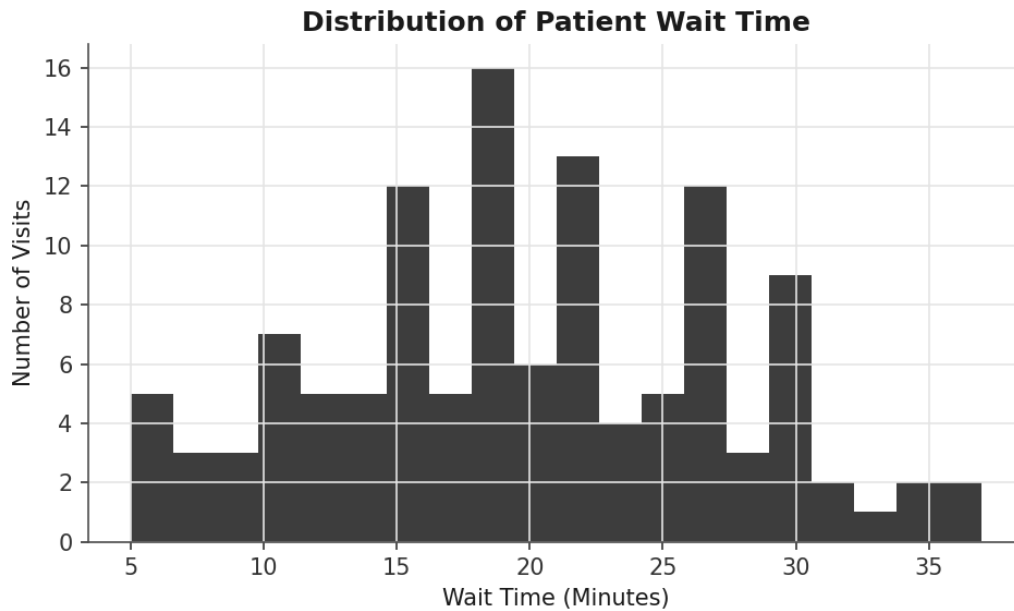


Figure 7: Distribution of Patient Wait Time

```

aes(x = factor(Patient_Rating), y = Number_of_Visits)
) +
geom_col() +
labs(
  title = "Distribution of Patient Satisfaction Ratings",
  x = "Patient Rating",
  y = "Number of Visits"
) +
theme_minimal()

```

15.2 Average Rating

```

average_rating <- mean(er_clean$Patient_Rating, na.rm = TRUE)
cat("Average patient rating:", round(average_rating, 2))

```

Average patient rating: 4.5

Interpretation

The rating distribution shows how patients evaluated their ER visits. The average rating provides an overall measure of patient satisfaction.

16. Patient Analysis

Number of Unique Patients

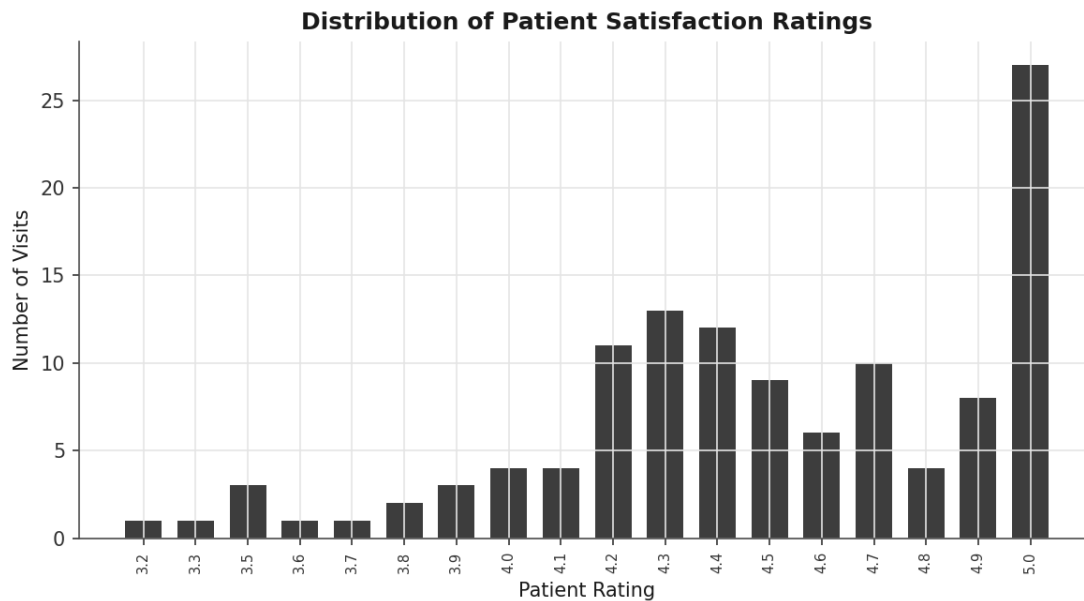


Figure 8: Distribution of Patient Satisfaction Ratings

```
number_of_patients <- length(unique(er_clean$Patient_ID))
cat("Number of unique patients:", number_of_patients)
```

```
## Number of unique patients: 103
```

Visits per Patient

```
patient_visits <- table(er_clean$Patient_ID)
patient_visits <- sort(patient_visits, decreasing = TRUE)
head(patient_visits, 10)
```

```
##
```

```
## PAT227 PAT423 PAT429 PAT464 PAT135 PAT185 PAT189 PAT200 PAT251 PAT330
```

```
##      3      3      3      3      2      2      2      2      2      2
```

Interpretation

The patient analysis identifies the number of unique patients and shows which patients had the greatest number of ER visits in the dataset. A small number of patients accounted for repeat visits, which may warrant further review of recurring cases.

17. Total Bill by Department

```
dept_bill <- aggregate(
  Total_Bill ~ Department_Type,
  data = er_clean,
  FUN = mean
```

```
)
names(dept_bill)[2] <- "Average_Total_Bill"
dept_bill <- dept_bill[order(-dept_bill$Average_Total_Bill), ]
dept_bill
```

```
##      Department_Type Average_Total_Bill
## 1      Cardiology      1051.93
## 6      Neurology      974.73
## 4 Gastroenterology      773.07
## 7      Orthopedics      724.67
## 3          ENT        583.40
## 5 General_Medicine      569.20
## 8      Pediatrics      495.80
## 2      Dermatology      397.20
```

```
ggplot(
  dept_bill,
  aes(x = reorder(Department_Type, Average_Total_Bill), y = Average_Total_Bill)
) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Average Total Bill by Department Type",
    x = "Department Type",
    y = "Average Total Bill"
  ) +
  theme_minimal()
```

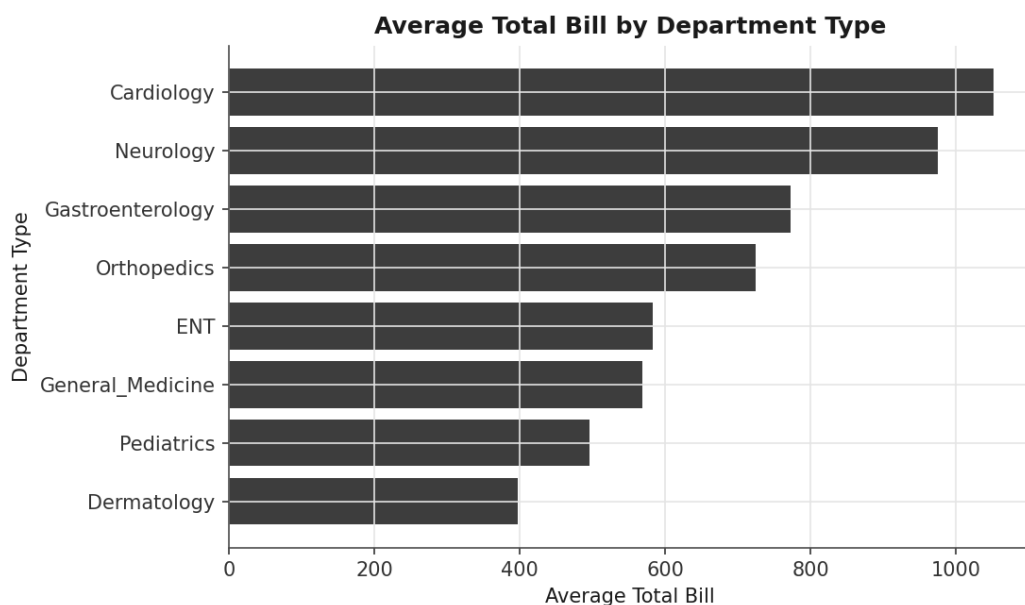


Figure 9: Average Total Bill by Department Type

Interpretation

The graph compares average billing across different department types. Cardiology and Neurology visits are associated with the highest average bills, consistent with the higher complexity of care in these departments, while Dermatology visits have the lowest average bills.

18. Treatment Time and Patient Rating

```
ggplot(  
  er_clean,  
  aes(x = Treatment_Time_Min, y = Patient_Rating)  
) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(  
    title = "Relationship Between Treatment Time and Patient Rating",  
    x = "Treatment Time (Minutes)",  
    y = "Patient Rating"  
  ) +  
  theme_minimal()
```

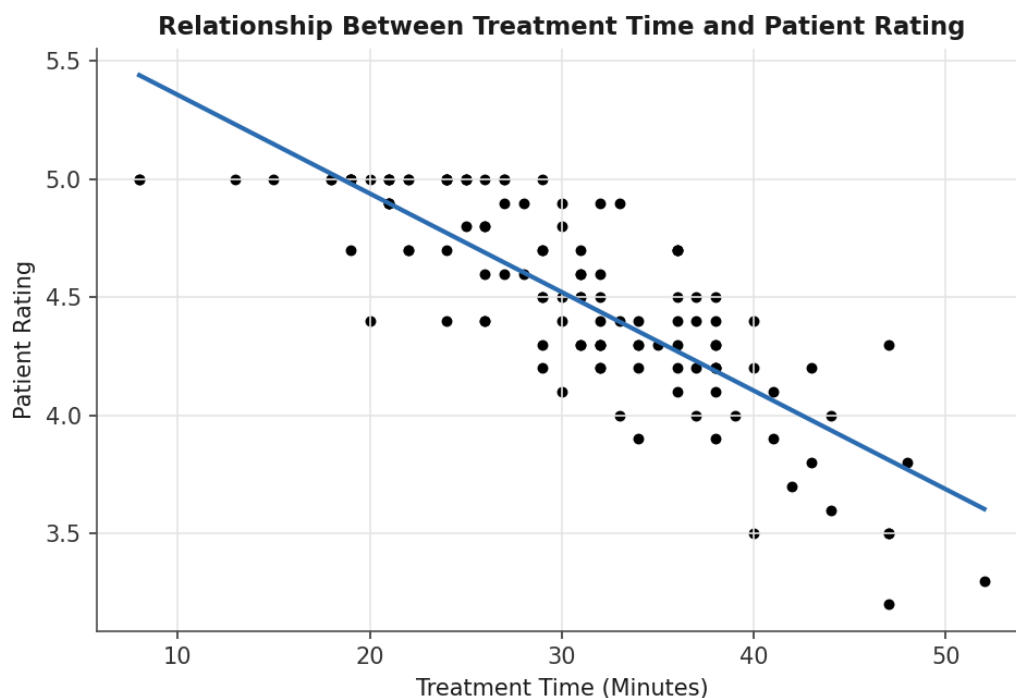


Figure 10: Relationship Between Treatment Time and Patient Rating

Interpretation

The scatter plot examines whether treatment time is associated with patient ratings. The downward trend line indicates that longer treatment times tend to be associated with lower satisfaction ratings. A relationship in the graph should not automatically be interpreted as causation.

19. Consultation Fee and Total Bill

```
ggplot(
  er_clean,
  aes(x = Consultation_Fee, y = Total_Bill)
) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Relationship Between Consultation Fee and Total Bill",
    x = "Consultation Fee",
    y = "Total Bill"
  ) +
  theme_minimal()
```

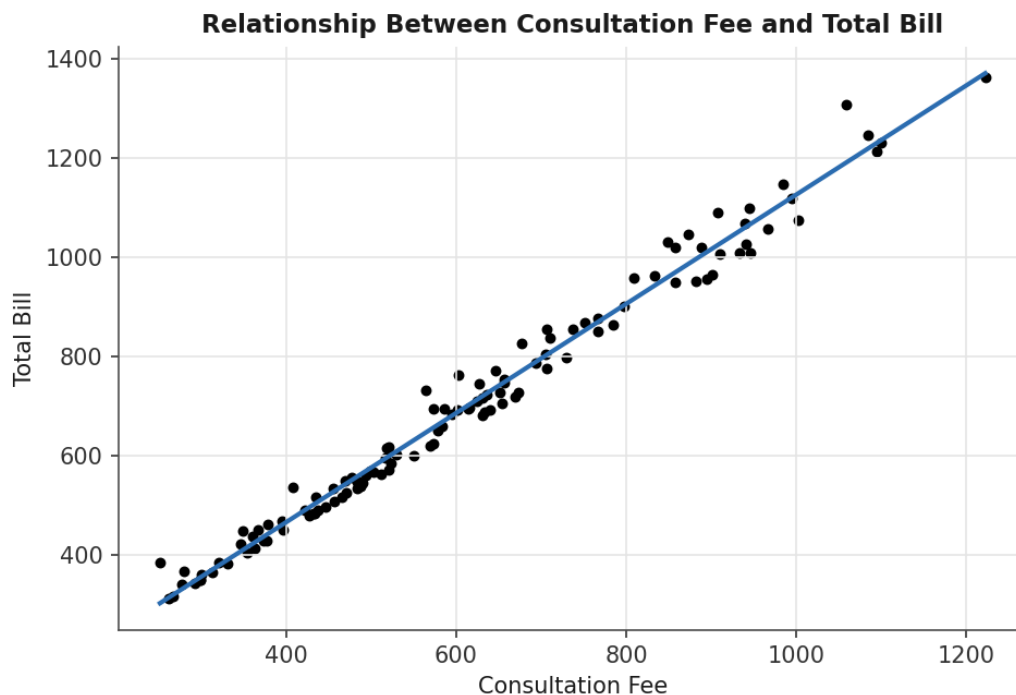


Figure 11: Relationship Between Consultation Fee and Total Bill

Interpretation

The graph examines the relationship between the consultation fee and the final total bill. Since the consultation fee is a major component of the total bill, a strong positive

relationship is expected and observed.

20. Correlation Analysis

Correlation measures the strength and direction of a linear relationship between numerical variables.

```
correlation_data <- er_clean[
  ,
  c(
    "Consultation_Fee",
    "Additional_Charges",
    "Total_Bill",
    "Wait_Time_Min",
    "Treatment_Time_Min",
    "Patient_Rating"
  )
]
correlation_matrix <- cor(correlation_data, use = "complete.obs")
round(correlation_matrix, 3)
```

```
##              Consultation_Fee Additional_Charges Total_Bill
## Consultation_Fee              1.000              0.584      0.992
## Additional_Charges            0.584              1.000      0.683
## Total_Bill                    0.992              0.683      1.000
## Wait_Time_Min                 0.530              0.413      0.542
## Treatment_Time_Min            0.621              0.468      0.633
## Patient_Rating                -0.537             -0.441     -0.552
##              Wait_Time_Min Treatment_Time_Min Patient_Rating
## Consultation_Fee            0.530              0.621     -0.537
## Additional_Charges          0.413              0.468     -0.441
## Total_Bill                   0.542              0.633     -0.552
## Wait_Time_Min                1.000              0.604     -0.580
## Treatment_Time_Min           0.604              1.000     -0.821
## Patient_Rating              -0.580             -0.821      1.000
```

Explanation

The `cor()` function calculates correlation coefficients. Correlation values range from -1 to +1:

- A value close to +1 indicates a strong positive relationship.
- A value close to -1 indicates a strong negative relationship.
- A value close to 0 indicates a weak linear relationship.

Interpretation

The correlation matrix helps identify relationships among consultation fees, additional charges, total billing, wait time, treatment time, and patient ratings. Treatment time

shows the strongest relationship with patient rating (-0.821), suggesting that visit duration is the single biggest driver of patient satisfaction in this dataset.

21. Total Visit Time Analysis

The total visit time represents the complete time a patient spends in the ER, from waiting through treatment.

```
total_time_stats <- c(
  Mean = mean(er_clean$Total_Visit_Time),
  Median = median(er_clean$Total_Visit_Time),
  Minimum = min(er_clean$Total_Visit_Time),
  Maximum = max(er_clean$Total_Visit_Time),
  Standard_Deviation = sd(er_clean$Total_Visit_Time)
)
total_time_stats
```

```
##              Mean              Median              Minimum
##          50.3750          50.0000          13.0000
##          Maximum Standard_Deviation
##          89.0000          14.0755
```

```
ggplot(er_clean, aes(x = Total_Visit_Time)) +
  geom_histogram(bins = 20) +
  labs(
    title = "Distribution of Total Visit Time",
    x = "Total Visit Time (Minutes)",
    y = "Number of Visits"
  ) +
  theme_minimal()
```

Interpretation

The graph shows the distribution of total time patients spend in the ER, from arrival through completion of treatment. This is useful for understanding overall ER efficiency.

22. Overall Project Summary

```
overall_summary <- data.frame(
  Total_Visits = nrow(er_clean),
  Unique_Patients = length(unique(er_clean$Patient_ID)),
  Unique_Hospitals = length(unique(er_clean$Hospital_Name)),
  Unique_Departments = length(unique(er_clean$Department_Type)),
  Average_Consultation_Fee = mean(er_clean$Consultation_Fee),
  Average_Total_Bill = mean(er_clean$Total_Bill),
  Average_Wait_Time = mean(er_clean$Wait_Time_Min),
  Average_Treatment_Time = mean(er_clean$Treatment_Time_Min),
```

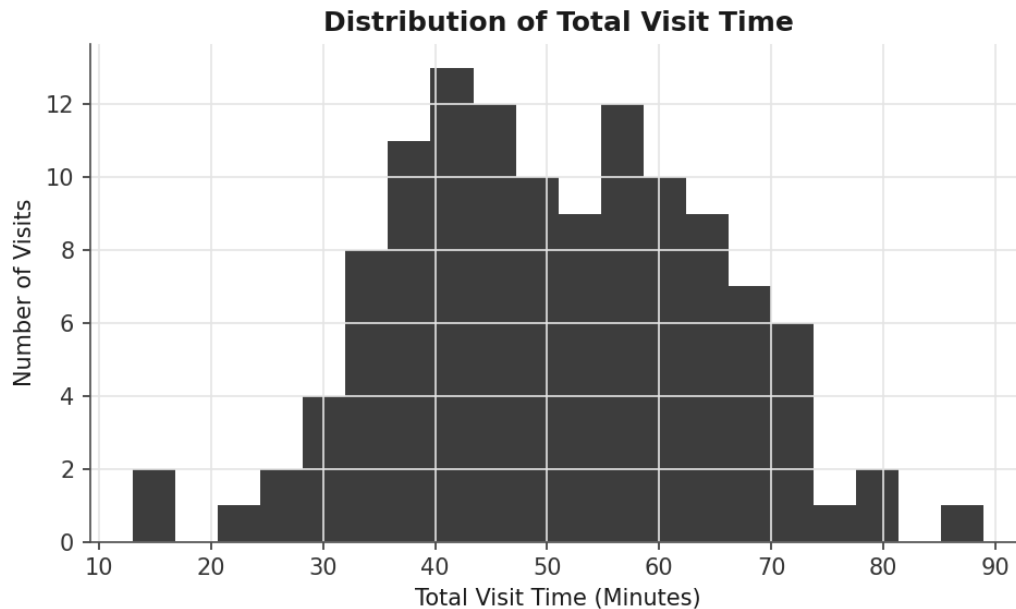


Figure 12: Distribution of Total Visit Time

```
Average_Rating = mean(er_clean$Patient_Rating)
)
overall_summary

## Total_Visits Unique_Patients Unique_Hospitals Unique_Departments
## 1          120           103             8             8
## Average_Consultation_Fee Average_Total_Bill Average_Wait_Time
## 1          609.81          696.25          19.73
## Average_Treatment_Time Average_Rating
## 1          30.65          4.495
```

23. Findings

Based on the statistical analysis and visualizations, the following findings were identified:

- The dataset contains detailed information about hospital ER visits.
- Consultation fees and total bills vary considerably across visits, largely driven by department type.
- Cardiology and Neurology visits are the most expensive on average, while Dermatology visits are the least expensive.
- Wait time and treatment time both vary noticeably among visits.
- Treatment time is the strongest driver of patient satisfaction (correlation of -0.821 with rating): longer treatment times are associated with lower ratings.
- Wait time also shows a moderate negative relationship with rating (-0.580).

- Consultation fee is almost perfectly correlated with total bill (0.992), as expected since it is the main component of the bill.
- The majority of visits (80%) were marked “Completed”, with smaller shares referred elsewhere or admitted for further care.
- A small number of patients had repeat ER visits within the dataset.

24. Business Recommendations

Based on the analysis, the following recommendations can be considered:

- Hospitals should prioritize reducing treatment time, since it has the strongest negative relationship with patient satisfaction.
- ER staff should monitor wait times closely and flag visits with unusually long waits for review.
- Hospitals can study high-cost departments (e.g., Cardiology, Neurology) to ensure billing transparency and staffing adequacy.
- Patient satisfaction ratings should be tracked by department to identify where service improvements are most needed.
- Hospitals should investigate patients with frequent repeat visits, as this may indicate unresolved health issues or gaps in follow-up care.
- Additional charges should be reviewed periodically to ensure they remain reasonable relative to consultation fees.

25. Conclusion

This project analyzed hospital ER visit data using R. The analysis included data importing, preprocessing, data cleaning, descriptive statistics, visit status analysis, department analysis, hospital analysis, cost analysis, wait-time and treatment-time analysis, patient rating analysis, patient analysis, and correlation analysis.

Various graphs were created to visualize important patterns and relationships within the dataset. Statistical measures were used to summarize the data and support the interpretation of results.

Overall, the project demonstrates how R can be used for practical data analysis and visualization in the healthcare industry. The results can help hospital administrators understand patient behavior, department demand, billing patterns, and ER performance, ultimately supporting better resource planning and improved patient care.

26. Explanation of R Functions Used

R Function	Purpose
<code>read.csv()</code>	Imports CSV data
<code>head()</code>	Displays first records
<code>nrow()</code>	Counts rows

R Function	Purpose
<code>ncol()</code>	Counts columns
<code>names()</code>	Displays column names
<code>str()</code>	Shows dataset structure
<code>summary()</code>	Provides summary statistics
<code>is.na()</code>	Identifies missing values
<code>duplicated()</code>	Identifies duplicate records
<code>unique()</code>	Finds unique values
<code>table()</code>	Creates frequency tables
<code>mean()</code>	Calculates average
<code>median()</code>	Calculates median
<code>min()</code>	Finds minimum
<code>max()</code>	Finds maximum
<code>sd()</code>	Calculates standard deviation
<code>aggregate()</code>	Calculates grouped statistics
<code>cor()</code>	Calculates correlation
<code>ggplot()</code>	Creates visualizations
<code>geom_col()</code>	Creates bar charts
<code>geom_histogram()</code>	Creates histograms
<code>geom_point()</code>	Creates scatter plots
<code>geom_smooth()</code>	Adds a trend line
<code>theme_minimal()</code>	Applies a clean graph theme

27. References

1. R Core Team. R: A Language and Environment for Statistical Computing.
2. Posit. RStudio IDE Documentation.
3. Hospital Emergency Room Visit dataset used for this project.